

A positive real period and its complete ES–Fable conductor return

Split-Zero programme: coordinate calculation

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Result SZ-20260921-005. This supplement gives the complete new root certificate, local inverse calculation and four-label return. All earlier proof sections cited by equation name are supplied in the complete cumulative `FABLE_TO_ORIGINAL_CONDUCTOR.tex` and its full supporting sources. Verified public programme sources have pinned links and exact proof locations in the bibliography. The quartet in the construction is a specified geometric member.

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1 A certified positive real period with exactly two vanishing factors

We construct an actual positive real period of the original geometric coefficient family at which the symbol has order exactly two. The point is defined by a unique root of the complete infinite series, with explicit enclosing inequalities and a real two-variable proof. The quartet, unit and conductor are those of PCL1–10 and WCF1–6 [5, 7]. The original four ES labels and their human sources remain FC1–15a and [3]. The geometric parameters chosen here are not asserted to be arithmetic zeta zeros.

1.1 The full original family and the exact real rectangle

Retain

$$\delta = \frac{1}{4}, \quad \gamma = 3, \quad \beta = \frac{143}{8}, \quad \eta = \frac{21025}{256}, \quad w = (\delta + i\gamma, \delta - i\gamma, -\delta + i\gamma, -\delta - i\gamma), \quad \rho_a = \frac{1}{2} + w_a. \quad (\text{RPZ1})$$

Here $V_{ar} = w_a^r$, $0 \leq r \leq 3$, and the entire original period matrix is

$$\mathcal{R}(z) = \sum_{n=0}^{\infty} R_n z^n, \quad (R_n)_{ar} = \sum_{\substack{p, h \geq 0 \\ 2p+4h=5n+r+1-a}} \frac{(\beta/3)^p \eta^h (-5)^{p+h-n} (a/5)_{p+h-n}}{p!h!}, \quad 1 \leq a \leq 4. \quad (\text{RPZ2})$$

The row index a in this sum is distinct from the actual unit below. Put $J_d = \text{diag}(1, -1, -1, 1)$ and $W(x) = I + ixJ_d$. For real x the original unit is $a = \alpha(1 + ix)$, with its retained nonzero real scale α , and $U = \alpha W(x) = \text{diag}(a, \bar{a}, \bar{a}, a)$. Its magnitude is $|\alpha|\sqrt{1+x^2}$. The common scalar cancels with its inverse only in the original matrix conjugation. For $x \in \mathbb{C} \setminus \{i, -i\}$ the same exact rational matrices give the analytic extension

$$\begin{aligned} W(x)^{-1} &= \frac{I - ixJ_d}{1 + x^2}, \quad D_j = \text{diag}(\zeta^{-j}, \zeta^{-2j}, \zeta^{-3j}, \zeta^{-4j}), \quad \zeta = e^{2\pi i/5}, \\ M_j(z, x) &= W(x)^{-1} V \mathcal{R}(z)^{-1} D_j \mathcal{R}(z) V^{-1} W(x), \\ f_j(\xi, z, x) &= Q_0(M_j(z, x)(e^{\rho_a \xi})_a), \quad Q_0(v) = v_1 v_4 - v_2 v_3, \\ d_j(z, x) &= f_j(0, z, x), \quad E_A(\xi, z, x) = \prod_{j=1}^4 f_j(\xi, z, x), \quad \mu_n(z, x) = \partial_{\xi}^n E_A(0, z, x). \end{aligned} \quad (\text{RPZ3})$$

The displayed extension follows by multiplication of W and its inverse; it is not restricted to UPP2's smaller disk. In particular the real rectangle below satisfies $1 + x^2 > 3$. The original PCL7–9 identity with the complete period conjugation, including the branch powers and Gamma ray factors, is unchanged.

The following terminating decimals are exact rational numbers:

$$\begin{aligned} z_c &= 0.02211947780866184513200791639903555836071244787217846841, \\ x_c &= -1.466817440463494183975448871845763045856183562396293859, \\ r_z &= r_x = 10^{-15}, \quad c_0 = (z_c, x_c), \\ X &= [z_c - r_z, z_c + r_z] \times [x_c - r_x, x_c + r_x] \subset \mathbb{R}^2. \end{aligned} \quad (\text{RPZ4})$$

There is a unique $(z_*, x_*) \in X$ with $d_1(z_*, x_*) = 0$. It lies in the interior, and defines the exact unit $a_* = \alpha(1 + ix_*)$ and the positive original period $u_* = 1/z_*$. The proof of this existence and uniqueness assertion is given next.

1.2 Full tails and derivatives used in the enclosing calculation

The entrywise estimate FPZ5–6, proved directly from the coefficients in RPZ2, is

$$|(R_n)_{ar}| \leq C_R(2n+2)\frac{T_R^n}{n!}, \quad K = 11/2, \quad C_R = 4K^3, \quad T_R = 6K^5/5. \quad (\text{RPZ5})$$

For completeness its inequalities use $L = p+h-n \geq 0$, $(a/5)_L \leq L!$, $n!L!/(p!h!) = \binom{p+h}{p}/\binom{p+h}{n} \leq 2^{p+h}$, $2(p+h) \leq 5n+3$, $2^{p+h} \leq 46^n$, and at most $2n+2$ contributing pairs. The inequalities $K^2 \geq 5\beta/3$ and $K^4 \geq 5\eta$ then give RPZ5 with every original sign retained in RPZ2. This majorant proves convergence of the matrix and all its derivative series on bounded sets.

Set $N = 650$, let ϱ_z be an outward upper bound for $|z|$ on the real interval in RPZ4, and put $q = T_R\varrho_z$. The complete omitted tails of the matrix and its first two derivatives are bounded by

$$\begin{aligned} \tau_0 &= \frac{C_R(2N+4)q^{N+1}}{(N+1)!(1-2q/(N+2))}, \\ \tau_1 &= \frac{C_R(2N+4)(N+1)T_Rq^N}{(N+1)!(1-3q/(N+1))}, \\ \tau_2 &= \frac{C_R(2N+4)N(N+1)T_R^2q^{N-1}}{(N+1)!(1-q(N+3)/(N(N+2)))}. \end{aligned} \quad (\text{RPZ6})$$

These follow by summing the first omitted majorant with the respective successive-term ratios bounded by $2q/(N+2)$, $3q/(N+1)$ and $q(N+3)/(N(N+2))$. All three bounds are less than one in the actual calculation. Simultaneous Horner recurrence evaluates the three finite sums; the respective tails are then added to each entry's real and imaginary interval. The computed upper bounds are

$$\tau_0 < 2.099 \cdot 10^{-161}, \quad \tau_1 < 9.485 \cdot 10^{-157}, \quad \tau_2 < 1.349 \cdot 10^{-152}. \quad (\text{RPZ7})$$

No omitted coefficient is set to zero. A centered Taylor integral further encloses each complete entry by

$$\begin{aligned} |\mathcal{R}_{ar}(z) - \mathcal{R}_{ar}(z_c) - \mathcal{R}'_{ar}(z_c)(z - z_c)| &\leq L_{ar}|z - z_c|^2/2, \\ |\mathcal{R}'_{ar}(z) - \mathcal{R}'_{ar}(z_c)| &\leq L_{ar}|z - z_c|, \end{aligned} \quad (\text{RPZ8})$$

where L_{ar} bounds the complete $|\mathcal{R}''_{ar}|$ on the interval. Every integration segment stays in that interval. Outward intervals enclosing the exact decimal center and radii are used throughout, so rounding the center is also included.

Let $A = \mathcal{R}^{-1}$ and $C_j = AD_j\mathcal{R}$. The derivative matrices are the literal products

$$\begin{aligned} (C_j)_z &= AD_j\mathcal{R}_z - A\mathcal{R}_zAD_j\mathcal{R}, \\ (C_j)_{zz} &= AD_j\mathcal{R}_{zz} - 2A\mathcal{R}_zAD_j\mathcal{R}_z + 2A\mathcal{R}_zA\mathcal{R}_zAD_j\mathcal{R} - A\mathcal{R}_{zz}AD_j\mathcal{R}, \\ (M_j)_{zh} &= W^{-1}V(C_j)_{zh}V^{-1}W \quad (h = 0, 1, 2), \\ (M_j)_x &= i[M_j, J_d]/(1+x^2), \quad (M_j)_{zx} = i[(M_j)_z, J_d]/(1+x^2), \\ (M_j)_{xx} &= -\{[[M_j, J_d], J_d] + 2ix[M_j, J_d]\}/(1+x^2)^2. \end{aligned} \quad (\text{RPZ9})$$

Differentiating $A\mathcal{R} = I$ gives $A_z = -A\mathcal{R}_zA$ and $A_{zz} = 2A\mathcal{R}_zA\mathcal{R}_zA - A\mathcal{R}_{zz}A$, which proves the first two lines with their stated matrix orders. For the phase derivative use $W^{-1}W_x = (xI + iJ_d)/(1+x^2)$; its scalar term commutes with M_j . Differentiation gives the remaining lines. With the original bilinear polar form $\mathfrak{b}(v, w) = v_1w_4 + v_4w_1 - v_2w_3 - v_3w_2$, $y = M_j\mathbf{1}$ and $h = M_j(\rho_a)_a$, the scalar derivatives are $d_j = \mathfrak{b}(y, y)/2$, $d_{j,z} = \mathfrak{b}(y, y_z)$, $d_{j,x} = \mathfrak{b}(y, y_x)$ and $f_{j,\xi} = \mathfrak{b}(y, h)$. The second derivatives used to enclose them follow by differentiating these products, exactly UPP6. In particular no Hermitian form replaces this bilinear form.

1.3 A completely specified real contraction

Define the real map $F(z, x) = (\Re d_1(z, x), \Im d_1(z, x))^T$. Use the fixed exact dyadic matrix

$$P = 2^{-64} \begin{pmatrix} 3218485790678941 & -57245565561083039 \\ -5877924768298298697 & -18277625187819728703 \end{pmatrix}, \quad T(y) = y - PF(y). \quad (\text{RPZ10})$$

The enclosing calculation is supplied as `certify_real_pair.py`. Its receipt is `REAL_PAIR_CERTIFICATE.json`. It uses Python-FLINT 0.9.0, 768-bit inclusion arithmetic and one worker. The arithmetic source is Fredrik Johansson's original Arb paper [8], Introduction and the sections on radii and precision. The present tail, derivative and fixed-point proof is given here, rather than attributed to that arithmetic reference.

The complete original matrix is invertible throughout X . Specifically, with A_c the dyadic midpoint matrix of the enclosing inverse at z_c , the full matrix enclosure gives $\|I - A_c \mathcal{R}(z)\|_\infty < 1.249 \cdot 10^{-11} < 1$. The convergent geometric matrix series in $I - A_c \mathcal{R}(z)$ inverts $A_c \mathcal{R}(z)$, so its two square factors, including $\mathcal{R}(z)$, are invertible. The inverse evaluation is enclosing. The determinant of RPZ10's exact matrix satisfies

$$-0.001161715807429 < \det P < -0.001161715807427. \quad (\text{RPZ11})$$

For reference, the full center Jacobian is enclosed about $\begin{pmatrix} 852.9039895042 & -2.671297323736 \\ -274.2864805121 & -0.150186872902 \end{pmatrix}$; all intervals, rather than this shortened display, enter the proof.

Let $\mathbf{J}$ enclose $DF(y)$ for all $y \in X$, let $\mathbf{E} = I - P\mathbf{J}$, and let $\boldsymbol{\eta}$ enclose $PF(c_0)$. For component indices $i, j \in \{z, x\}$ put

$$q_i = \frac{\sum_j \sup |\mathbf{E}_{ij}| r_j}{r_i}, \quad b_i = \frac{\sup |\boldsymbol{\eta}_i| + \sum_j \sup |\mathbf{E}_{ij}| r_j}{r_i}. \quad (\text{RPZ12})$$

The complete enclosures verify the strict rational inequalities

$$q_z, b_z < 4.578 \cdot 10^{-9}, \quad q_x, b_x < 1.829 \cdot 10^{-6} < 1. \quad (\text{RPZ13})$$

All operations defining the enclosures, the full defect matrix, center residual, exact dyadic P , and tail bounds are retained in the executable and receipt. They are sufficient to replay each displayed strict inequality without any numerical root assumption.

Here is the existence proof. Equip $\mathbb{R}^2$ with $\|v\|_r = \max_i |v_i|/r_i$. Integration of $DT = I - P D F$ along the segment in the convex rectangle X gives $\|T(y) - T(y')\|_r \leq q \|y - y'\|_r$, where $q = \max_i q_i < 1$. Also $T(c_0) - c_0 = -PF(c_0)$, so the same integration gives $|T(y)_i - (c_0)_i| \leq b_i r_i < r_i$. Thus T maps X into its interior and contracts it. Start $y_0 = c_0$ and set $y_{n+1} = T(y_n)$. Induction keeps every iterate in X , and

$$\|y_m - y_n\|_r \leq \frac{q^n}{1 - q} \|y_1 - y_0\|_r \quad (m > n). \quad (\text{RPZ14})$$

This follows by summing the geometric bounds on successive differences. Hence the iterates form a Cauchy sequence. Completeness of $\mathbb{R}^2$ and closedness of X give a limit $y_* \in X$. Continuity gives $T(y_*) = y_*$, and RPZ11 gives $F(y_*) = 0$. Any two fixed points have distance at most q times their distance, so they coincide. Every zero of F is a fixed point. Finally $y_* = T(y_*)$ lies in the interior by the strict self-map bound. This proves every existence and uniqueness statement after RPZ4.

1.4 The exact pair, moments and nonzero inverse coefficient

On the entire real rectangle the factor enclosures include

$$\begin{aligned} d_{1,z} &\in [852.90399 \pm 1.90 \cdot 10^{-6}] + i[-274.28648 \pm 1.91 \cdot 10^{-6}], \\ f_{1,\xi} &\in [17.1783460 \pm 3.61 \cdot 10^{-8}] + i[-6.8131052 \pm 4.44 \cdot 10^{-8}], \\ d_2 &\in [0.05410876 \pm 1.84 \cdot 10^{-9}] + i[-0.31535702 \pm 5.02 \cdot 10^{-9}]. \end{aligned} \quad (\text{RPZ15})$$

They exclude zero. Let J_c exchange the first two and the last two primary coordinates; it is distinct from J_d in RPZ3. The real coefficient matrix in RPZ2, $\bar{V} = J_c V$, $\bar{U} = J_c U J_c$, and $Q_0(J_c v) = -Q_0(v)$ give

$$f_{5-j}(\xi, z, x_*) = -\overline{f_j(\bar{\xi}, \bar{z}, x_*)}, \quad E_A = GG^\#, \quad G = f_1 f_2, \quad G^\#(\xi, z) = \overline{G(\bar{\xi}, \bar{z})}. \quad (\text{RPZ16})$$

This is the earlier OFM8–11 conjugation and positivity identity [5, 1], with its full two-variable use derived in RPCJ1–4 below. It fixes the original actual unit. At the certified point exactly $d_1 = d_4 = 0$ and $d_2, d_3 \neq 0$. Writing $A = G_z(0, z_*, x_*)$ and $B = G_\xi(0, z_*, x_*)$, RPZ15 gives $A, B \neq 0$. Consequently the full analytic products give

$$\begin{aligned} \mu_0(z, x_*) &= |A|^2(z - z_*)^2 + O((z - z_*)^3), \\ \mu_0(z_*, x_*) &= \mu_1(z_*, x_*) = 0, \quad \mu_2(z_*, x_*) = 2|B|^2 > 0. \end{aligned} \quad (\text{RPZ17})$$

Thus the period zero has multiplicity exactly two and the original symbol has order exactly two, in the original coordinates. Every nontrivial orbit quadric has nonzero restriction to the exponential curve: for $j = 1, 4$ its first derivative is nonzero, and for $j = 2, 3$ its value is nonzero. Since $Q_0(t(\xi)) = 0$ identically, none is associated to Q_0 . If two orbit quadrics were associated, applying the inverse cyclic action would associate a nontrivial member to Q_0 . Therefore all five orbit quadrics are distinct. This does not assert pairwise coprimality after every unrelated section.

The full original derivatives, not freely chosen jets, satisfy

$$\begin{aligned} A &\in [-40.34859 \pm 1.87 \cdot 10^{-6}] + i[-283.81057 \pm 6.15 \cdot 10^{-6}], \\ B &\in [-1.2190616 \pm 5.30 \cdot 10^{-8}] + i[-5.7859608 \pm 6.51 \cdot 10^{-8}], \\ c &= |A|^2 \in [82176.45 \pm 0.00525], \quad b = |B|^2 \in [34.963453 \pm 4.49 \cdot 10^{-7}], \\ R &= 2\Re(A\bar{B}) \in [3382.6084 \pm 0.0000642], \quad 2cb - R^2 \in [-5695695 \pm 0.547]. \end{aligned} \quad (\text{RPZ18})$$

In particular $R(2cb - R^2) \neq 0$. The complete original moment enclosures through order five are supplied by RPE2–3 below; their exact finite derivative formula retains all four factors. At this actual point the generic inverse coefficient of RPCJ15–16 is nonzero. Hence the fixed old degree-three conductor has inverse singular exponents $(5, 3, 0, 0)$ and inverse exterior exponents $(5, 8, 8, 8)$, with every original Gamma constant calculated in RPCJ20–22 and RPE5–9. The exterior rank-one leading coefficient before its original factor ρ_3/ρ_0 is

$$\frac{|R(2cb - R^2)|}{c^4} \in [4.224828 \cdot 10^{-10} \pm 7.04 \cdot 10^{-17}], \quad u_* \in [45.20902385898 \pm 3.31 \cdot 10^{-12}]. \quad (\text{RPZ19})$$

The source order change, four-label return, and both lost directions are calculated below on their full original spaces. This positive real geometric period is an actual obstruction to a finite uniform inverse bound on the whole geometric family. Locating the arithmetic quartet and its restricted period/unit locus requires their exact arithmetic map; the present construction makes no identification of that locus with this explicitly certified member.

2 The exact real-period conjugate-pair collision and its two inverse spectra

This calculation retains the original unit, quartet, periods and all Gamma metrics of PCL1–10, OFM1–17, WCF1–6 and GFC1–21 [5, 4, 1, 7]. It calculates the exact local matrix on the locus of simple conjugate-pair collisions defined below. The RPZ full-series contraction proof supplies an actual positive real-period point on this locus, and RPE1–26 specializes every constant and four-label map there. The general calculation below also determines the complete exceptional coefficient locus.

2.1 Joint conjugation in both original variables

Retain $a = \alpha + i\chi \neq 0$ with real α, χ , the unchanged $U = \text{diag}(a, \bar{a}, \bar{a}, a)$, and the original ordered quartet in V . Let J exchange primary indices 1, 2 and 3, 4. For $z = 1/u$ and the exponential-curve coordinate ξ put

$$L(z) = U^{-1}V\mathcal{R}(z)^{-1}, \quad M_j(z) = L(z)D_5^{-j}L(z)^{-1}, \quad f_j(\xi, z) = Q_0(M_j(z)t(\xi)). \quad (\text{RPCJ1})$$

Every PCL2 coefficient is real at the original real δ, γ , so $\overline{\mathcal{R}(\bar{z})} = \mathcal{R}(z)$. The original labels and unit give $\bar{V} = JV$, $\bar{U} = JUJ$, $\overline{t(\bar{\xi})} = Jt(\xi)$, and $Q_0(Jv) = -Q_0(v)$. Therefore

$$\overline{L(\bar{z})} = JL(z), \quad \overline{M_j(\bar{z})} = JM_{5-j}(z)J, \quad \boxed{f_{5-j}(\xi, z) = -\overline{f_j(\bar{\xi}, \bar{z})}}. \quad (\text{RPCJ2})$$

To verify the first equality multiply $\bar{U}^{-1}\bar{V} = (JU^{-1}J)(JV) = JU^{-1}V$. Conjugating the diagonal D_5^{-j} gives D_5^j and proves the second equality. Substituting the conjugated original t and then using the exact sign in $Q_0(Jv)$ proves the final equality. This is the two-variable form of the earlier OFM8–10 identity; it keeps the original unit fixed rather than replacing its phase.

For $h^\#(\xi, z) = \overline{h(\bar{\xi}, \bar{z})}$, set $G = f_1 f_2$. The two minus signs in RPCJ2 cancel, giving

$$E_A(\xi, z) = G(\xi, z)G^\#(\xi, z), \quad E_A(\xi, z) = |G(\xi, z)|^2 \geq 0 \quad (\xi, z \in \mathbb{R}). \quad (\text{RPCJ3})$$

At every real period, the full moment formula is

$$\mu_n(z) = \sum_{r=0}^n \binom{n}{r} \partial_\xi^r G(0, z) \overline{\partial_\xi^{n-r} G(0, z)}. \quad (\text{RPCJ4})$$

Each derivative of G is itself the complete original factor sum $\partial_\xi^r G = \sum_{h=0}^r \binom{r}{h} (\partial_\xi^h f_1)(\partial_\xi^{r-h} f_2)$. Thus RPCJ4 retains every original moment and repeated contribution. The finite nine-frequency test and order bound from PZ12 and OFM2–5 apply with their existing provenance [6, 1]: when the symbol is nonzero, $n_4 = n_1$, $n_3 = n_2$, and its order is $v = 2(n_1 + n_2) \leq 32$ with positive leading moment. An identically zero factor instead gives an identically zero symbol, as in GFC13–14; no finite order is assigned in that case.

2.2 The exact simple-pair locus and every moment there

Define the local locus under study by an original real $z_0 \neq 0$ at which exactly one of $f_1(0, z_0)$ and $f_2(0, z_0)$ vanishes, the other is nonzero, and the vanishing factor has nonzero first derivatives with respect to both z and ξ at $(0, z_0)$. RPCJ2 forces its conjugate partner to vanish with the same simple orders, while the other partner is nonzero. This definition retains all four factors; it is not a certificate that this locus is nonempty. Write the complete actual jets

$$\begin{aligned} A &= G_z(0, z_0) \neq 0, & B &= G_\xi(0, z_0) \neq 0, \\ C &= G_{zz}(0, z_0), & H &= G_{\xi z}(0, z_0), & F &= G_{\xi\xi}(0, z_0), & \tau &= z - z_0. \end{aligned} \quad (\text{RPCJ5})$$

For example when $f_1(0, z_0) = 0$ and $b = f_2(0, z_0) \neq 0$, these are the full expressions

$$\begin{aligned} A &= b f_{1,z}, & B &= b f_{1,\xi}, \\ C &= b f_{1,zz} + 2 f_{1,z} f_{2,z}, \\ H &= b f_{1,\xi z} + f_{1,z} f_{2,\xi} + f_{1,\xi} f_{2,z}, \\ F &= b f_{1,\xi\xi} + 2 f_{1,\xi} f_{2,\xi}. \end{aligned} \quad (\text{RPCJ6})$$

All derivatives on the right are evaluated at $(0, z_0)$ in the original functions RPCJ1. When f_2 is the vanishing factor the same product derivatives exchange the two displayed labels.

The complete mixed moment formula, valid for every $m, n \geq 0$ at this real period, is

$$\partial_z^m \mu_n(z_0) = \sum_{r=0}^n \sum_{h=0}^m \binom{n}{r} \binom{m}{h} \partial_\xi^r \partial_z^h G(0, z_0) \overline{\partial_\xi^{n-r} \partial_z^{m-h} G(0, z_0)}. \quad (\text{RPCJ7})$$

It follows by differentiating the exact analytic identity RPCJ3; complex conjugation of the coefficients is justified by the reality of z_0 , not by applying conjugation to a complex independent variable. The leading cases are

$$\begin{aligned} \mu_0(z) &= |A|^2 \tau^2 + \text{Re}(C\bar{A})\tau^3 + O(\tau^4), \\ \mu_1(z) &= 2 \text{Re}(A\bar{B})\tau + [\text{Re}(B\bar{C}) + 2 \text{Re}(H\bar{A})]\tau^2 + O(\tau^3), \\ \mu_2(z_0) &= 2|B|^2 > 0, \quad \mu_2'(z_0) = 2 \text{Re}(F\bar{A}) + 4 \text{Re}(H\bar{B}), \quad \mu_3(z_0) = 6 \text{Re}(F\bar{B}). \end{aligned} \quad (\text{RPCJ8})$$

In particular the actual period-zero multiplicity is $\nu = 2$, while the actual exponential-symbol order is $v = 2$. Both are calculated from nonzero original derivatives, not identified by their names. The higher moments are all specified by RPCJ7.

2.3 The exact four-dimensional weighted inverse

Now use the unchanged old cutoff $D = 3$, with $s \in \{1, a_{\text{amp}}\}$, the original centers $a_{\text{amp}}/2$ and $a_{\text{amp}}/2 - 4$, the complete mass $M_s = (2\pi)^{s/2}$, and $\rho_m = \sqrt{m!/(s/2)_m}$, including $\rho_0 = 1$. These are precisely the physical norms of GFC4; the following matrix is in their original orthonormal frames. To avoid confusing $G = f_1 f_2$ with the conductor series, retain

$$\begin{aligned} g(t, z) &= e^{-4\omega(t)} E_A(\omega(t), z), \quad \omega(t) = -i \arctan t, \\ B_s(z) &= \text{diag}(\rho_0, \dots, \rho_3)^{-1} T_3(g) \text{diag}(\rho_0, \dots, \rho_3). \end{aligned} \quad (\text{RPCJ9})$$

Its four coefficients, with every center correction, are

$$\begin{aligned} g_0 &= \mu_0, \quad g_1 = -i(\mu_1 - 4\mu_0), \\ g_2 &= -\mu_2/2 + 4\mu_1 - 8\mu_0, \\ g_3 &= i(\mu_3/6 - 2\mu_2 + 25\mu_1/3 - 12\mu_0). \end{aligned} \quad (\text{RPCJ10})$$

Expanding $\omega = -it + it^3/3 + O(t^5)$ and the exact exponential proves these four coefficients directly. Define the following actual constants, including their signs:

$$\begin{aligned} c &= |A|^2 > 0, \quad R = 2 \text{Re}(A\bar{B}), \quad d = -iR, \quad e = -|B|^2 < 0, \\ c_3 &= \text{Re}(C\bar{A}), \quad d_2 = -i[\text{Re}(B\bar{C}) + 2 \text{Re}(H\bar{A}) - 4c], \\ e_1 &= -\text{Re}(F\bar{A}) - 2 \text{Re}(H\bar{B}) + 4R, \quad f = i[\text{Re}(F\bar{B}) - 4|B|^2]. \end{aligned} \quad (\text{RPCJ11})$$

Equations RPCJ8–10 give

$$g_0 = c\tau^2 + c_3\tau^3 + O(\tau^4), \quad g_1 = d\tau + d_2\tau^2 + O(\tau^3), \quad g_2 = e + e_1\tau + O(\tau^2), \quad g_3 = f + O(\tau). \quad (\text{RPCJ12})$$

At $\tau = 0$ the nonzero matrix block is

$$C_s = \begin{pmatrix} e\rho_2/\rho_0 & f\rho_3/\rho_0 \\ 0 & e\rho_3/\rho_1 \end{pmatrix}. \quad (\text{RPCJ13})$$

Its columns are source indices 2, 3 and its rows are target indices 0, 1 in the displayed coordinate matrix; equivalently the original map sends source degrees 2, 3 to target degrees 0, 1. The matrix C_s is invertible because $e \neq 0$. Thus $B_s(z_0)$ has rank two, and its two positive forward singular values are exactly the two singular values of this complete block.

The inverse series through degree three is exactly

$$\begin{aligned} h_0 &= \frac{1}{g_0}, \quad h_1 = -\frac{g_1}{g_0^2}, \\ h_2 &= \frac{g_1^2 - g_0 g_2}{g_0^3}, \quad h_3 = \frac{-g_1^3 + 2g_0 g_1 g_2 - g_0^2 g_3}{g_0^4}, \\ B_s^{-1} &= \text{diag}(\rho)^{-1} T_3(h) \text{diag}(\rho). \end{aligned} \tag{RPCJ14}$$

Multiplication of the two truncated series proves all entries. Set

$$\Delta = d(d^2 - 2ce), \quad L = \frac{d^2 - ce}{c^3}. \tag{RPCJ15}$$

If $\Delta \neq 0$, the only inverse entry with order five is $h_3 \rho_3 / \rho_0$. Its leading coefficient is $-\Delta \rho_3 / (c^4 \rho_0)$. Every other entry has order at most four by RPCJ12–14. It follows that the largest inverse singular value has the exact limit

$$\lim_{z \rightarrow z_0} |z - z_0|^5 \|B_s(z)^{-1}\| = \frac{|\Delta| \rho_3}{c^4 \rho_0} > 0. \tag{RPCJ16}$$

The exceptional coefficient locus $\Delta = 0$ is also completely determined. Put

$$\theta = \text{Re}(A\bar{B}) / (|A||B|) \in [-1, 1].$$

This quotient is an exact auxiliary value of the two original derivatives; their magnitudes remain in RPCJ11 and every constant. Then $\Delta = 0$ exactly when

$$\theta = 0 \quad \text{or} \quad \theta^2 = 1/2. \tag{RPCJ17}$$

Indeed $d = -2i|A||B|\theta$ and $ce = -|A|^2|B|^2$. On this locus d^2 equals either 0 or $2ce$, so $d^2 - ce$ equals $-ce$ or ce and is nonzero. Thus $L \neq 0$ in all exceptional cases. The complete next coefficient of the top-right inverse entry is

$$K = \frac{2cde_1 + 2c_3de + (2ce - 3d^2)d_2 - c^2f}{c^4}. \tag{RPCJ18}$$

To prove it, the coefficient of τ^4 in the numerator of h_3 in RPCJ14 is exactly the displayed numerator, whereas its τ^3 coefficient is $-\Delta = 0$. The denominator has leading coefficient $c^4 \tau^8$, so no denominator correction contributes at order τ^{-4} . Consequently the full leading inverse block is

$$N_s = \begin{pmatrix} L\rho_2/\rho_0 & K\rho_3/\rho_0 \\ 0 & L\rho_3/\rho_1 \end{pmatrix}, \quad \lim_{z \rightarrow z_0} |z - z_0|^4 \|B_s(z)^{-1}\| = \|N_s\| > 0 \quad (\Delta = 0). \tag{RPCJ19}$$

For this limit, multiplication by τ^4 makes every other inverse entry tend to zero, while these four block entries have the stated limits. Continuity of the operator norm proves RPCJ19 for complex as well as real approaches to the original real period.

2.4 Every exterior rank and every inverse singular value

Independently of Δ , the exact complementary-exterior identity GFC11 and $\det B_s = \mu_0^4$ prove

$$\begin{aligned} \lim_{z \rightarrow z_0} |z - z_0|^8 \left\| \bigwedge^2 B_s(z)^{-1} \right\| &= \frac{|e|^2 \rho_2 \rho_3}{c^4 \rho_0 \rho_1}, \\ \lim_{z \rightarrow z_0} |z - z_0|^8 \left\| \bigwedge^3 B_s(z)^{-1} \right\| &= \frac{\|C_s\|}{c^4}, \\ \lim_{z \rightarrow z_0} |z - z_0|^8 \left\| \bigwedge^4 B_s(z)^{-1} \right\| &= c^{-4}. \end{aligned} \tag{RPCJ20}$$

The first numerator is $|\det C_s|$, because the only nonzero second exterior singular value of the rank-two limiting matrix is the product of its two positive singular values. The next numerator is its largest singular value $\|C_s\|$. All displayed constants are strictly positive.

For an explicit evaluation of either block norm, if $T = \begin{pmatrix} u & v \\ 0 & w \end{pmatrix}$, put $S_T = |u|^2 + |v|^2 + |w|^2$ and $P_T = |uw|^2$. The two singular values are

$$\sigma_{\pm}(T) = \left(\frac{S_T \pm \sqrt{S_T^2 - 4P_T}}{2} \right)^{1/2}, \quad \|T\| = \sigma_+(T). \quad (\text{RPCJ21})$$

This follows from the trace and determinant of T^*T and keeps both exact off-diagonal coefficients in RPCJ13 and RPCJ19.

Let the inverse singular values be in descending order. For $\Delta \neq 0$, their pole exponents and leading constants are

$$(5, 3, 0, 0), \quad \left(\frac{|\Delta|\rho_3}{c^4\rho_0}, \quad \frac{|e|^2\rho_2}{|\Delta|\rho_1}, \quad \frac{1}{\sigma_-(C_s)}, \quad \frac{1}{\sigma_+(C_s)} \right). \quad (\text{RPCJ22})$$

The first two constants follow by dividing the first line of RPCJ20 by RPCJ16. The last two follow from the convergence of the two nonzero forward singular values to $\sigma_{\pm}(C_s)$. For $\Delta = 0$, the complete answer is instead

$$(4, 4, 0, 0), \quad \left(\sigma_+(N_s), \sigma_-(N_s), \frac{1}{\sigma_-(C_s)}, \frac{1}{\sigma_+(C_s)} \right). \quad (\text{RPCJ23})$$

The limiting rank-two inverse block in RPCJ19 is invertible, so both first constants are nonzero. As a check retaining all weights, on this locus $|L| = |e|/c^2$ and hence $|\det N_s| = |\det C_s|/c^4$, exactly RPCJ20's first coefficient. Thus the exterior exponents are $(5, 8, 8, 8)$ or $(4, 8, 8, 8)$, respectively. Nonzero first z and ξ derivatives alone do not select between these two calculated possibilities.

At the collision the order-adjusted map is the exact $v = 2$ map $\mathcal{P}_5/\mathcal{P}_1 \rightarrow \mathcal{P}_3$ with its original Gamma quotient norm and diagonal $(-i)^2\mu_2\rho_{j+2}/(2\rho_j)$, $0 \leq j \leq 3$. It is invertible because $\mu_2 = 2|B|^2 > 0$. This distinct domain is retained alongside the singular fixed old map; no quotient change is used to erase the fixed-space poles.

Finally $u_0 = 1/z_0$ is real and nonzero, and $z - z_0 = -(u - u_0)/(uu_0)$ exactly. Each singular or exterior pole of exponent p above therefore has the same exponent in the original u coordinate and its displayed constant multiplied by $|u_0|^{2p}$. This specifies every original coordinate constant without selecting a candidate real period.

3 The certified positive real period: complete four-label return and correction

This section applies the full RPCJ calculation to the actual original real-period point established by the accompanying contraction proof. The ordered ES-Fable labels and polynomial retain FC5–6, FC15a and FC31–35 and the original 488-page reader's provenance [3, 1]; the conductor and every norm retain WCF1–6 [7]. The conjugation and real-axis positivity used here are the existing OFM8–11 and earlier PCL15–16 identities [5, 1], not a new parity or positivity theorem.

The exact point (z_*, x_*) is the unique real solution of $f_1(0, z, x) = 0$ in the closed rectangle of radius 10^{-15} in each coordinate about the exact rational center

$$\begin{aligned} z_c &= 0.02211947780866184513200791639903555836071244787217846841, \\ x_c &= -1.466817440463494183975448871845763045856183562396293859. \end{aligned} \quad (\text{RPE1})$$

The original parameters are $\delta = 1/4$, $\gamma = 3$, $\beta = 143/8$, $\eta = 21025/256$; the exact unit is $a = \alpha(1 + ix_*)$ for its retained real nonzero scale α , and the original period is $u_* = 1/z_* > 0$. The scale cancels with its inverse only in the original conjugated operator, as in PCL9. Every remaining unit entry is kept there. The full-series certificate proves $f_1 = f_4 = 0$, $f_2, f_3 \neq 0$, and nonzero z and ξ derivatives of f_1 . Thus RPCJ8 gives exactly $\nu = 2$ and $v = 2$ at this point. The same certificate proves the five-orbit property at this point. The two vanishing factors are nonzero functions of ξ because their first ξ derivatives are nonzero; the other two factors have nonzero value. If a nontrivial conjugated quadric were associated to Q_0 , its restriction to the original exponential curve would vanish identically, contrary to those statements. Association of two distinct orbit members would reduce to such an association by their inverse group action. Hence all five original quadrics are distinct.

All moments used below are the actual ones at this certified point. For a completely explicit finite expression, let $M_j = M_j(z_*, x_*)$ be the original matrix, $v_{j,n} = M_j(\rho_a^n)_a$, and use the literal polar form $\mathcal{B}(v, w) = v_1 w_4 + v_4 w_1 - v_2 w_3 - v_3 w_2$. Then

$$a_{j,n} = \frac{1}{2} \sum_{r=0}^n \binom{n}{r} \mathcal{B}(v_{j,r}, v_{j,n-r}), \quad G_n = \sum_{r=0}^n \binom{n}{r} a_{1,r} a_{2,n-r}, \quad \mu_n = \sum_{r=0}^n \binom{n}{r} G_r \bar{G}_{n-r}. \quad (\text{RPE2})$$

The factor $1/2$ retains the original $Q_0(v) = \mathcal{B}(v, v)/2$. These formulas evaluate every needed derivative of the full original period and factor matrices, including all four multiplicities through their exact conjugation. The certificate supplies the real enclosures

$$\begin{aligned} \mu_0 &= \mu_1 = 0, & \mu_2 &\in [69.926906 \pm 8.99 \cdot 10^{-7}], \\ \mu_3 &\in [-11161.6109 \pm 8.96 \cdot 10^{-5}], & \mu_4 &\in [576883.78 \pm 6.81 \cdot 10^{-3}], & \mu_5 &\in [5.254785 \cdot 10^6 \pm 0.252]. \end{aligned} \quad (\text{RPE3})$$

The zero values here are exact consequences of the certified root and RPCJ3, not the rounding of an interval through zero. The real parts of the other enclosures are valid because RPCJ4 proves the exact moments real at this real period.

3.1 Every inverse constant at the certified period

Retain RPCJ11's exact constants $c = |G_z|^2$, $R = 2\text{Re}(G_z \bar{G}_\xi)$, $d = -iR$, $e = -\mu_2/2$, and $f = i(\mu_3/6 - 2\mu_2)$. The full-series certificate gives

$$\begin{aligned} c &\in [82176.45 \pm 0.00526], & R &\in [3382.6084 \pm 0.0000643], \\ 2c|G_\xi|^2 - R^2 &\in [-5695695 \pm 0.547], & |\Delta| &= |d(d^2 - 2ce)| \in [1.926631 \cdot 10^{10} \pm 6470]. \end{aligned} \quad (\text{RPE4})$$

In particular $\Delta \neq 0$ at this actual point; RPCJ22's generic case therefore occurs. For every retained original s , put

$$C_s = \begin{pmatrix} -\frac{\mu_2}{2} \frac{\rho_2}{\rho_0} & i(\mu_3/6 - 2\mu_2) \frac{\rho_3}{\rho_0} \\ 0 & -\frac{\mu_2}{2} \frac{\rho_3}{\rho_1} \end{pmatrix}, \quad (p_1, p_2, p_3, p_4) = (5, 8, 8, 8). \quad (\text{RPE5})$$

The complete exact exterior constants are

$$\lim_{z \rightarrow z_*} |z - z_*|^{p_r} \left\| \bigwedge^r B_{3,s}(z)^{-1} \right\| = \left(\frac{|\Delta| \rho_3}{c^4 \rho_0}, \quad \frac{\mu_2^2 \rho_2 \rho_3}{4c^4 \rho_0 \rho_1}, \quad \frac{\|C_s\|}{c^4}, \quad c^{-4} \right)_r. \quad (\text{RPE6})$$

The block norm is the explicit expression RPCJ21 applied to the three entries in RPE5. This retains the actual μ_3 rather than bounding or deleting its off-diagonal contribution. The inverse singular exponents are $(5, 3, 0, 0)$ and their constants are exactly

$$\left(\frac{|\Delta| \rho_3}{c^4 \rho_0}, \quad \frac{\mu_2^2 \rho_2}{4|\Delta| \rho_1}, \quad \sigma_-(C_s)^{-1}, \quad \sigma_+(C_s)^{-1} \right). \quad (\text{RPE7})$$

At $s = 1$, the original weights give $(\rho_0, \rho_1, \rho_2, \rho_3) = (1, \sqrt{2}, \sqrt{8/3}, \sqrt{16/5})$. Propagation of the saved certificate enclosures through these exact formulas gives all four exterior constants, respectively,

$$\begin{aligned} & [7.55760 \cdot 10^{-10} \pm 3.35 \cdot 10^{-16}], \quad [5.53711 \cdot 10^{-17} \pm 5.87 \cdot 10^{-23}], \\ & [7.84748 \cdot 10^{-17} \pm 6.86 \cdot 10^{-23}], \quad [2.192858 \cdot 10^{-20} \pm 6.28 \cdot 10^{-27}]. \end{aligned} \quad (\text{RPE8})$$

The last two inverse singular limits at this order are in

$$[1.4172517 \pm 7.40 \cdot 10^{-8}], \quad [0.00027943449 \pm 5.32 \cdot 10^{-12}].$$

The second inverse singular leading constant is in $[7.32655 \cdot 10^{-8} \pm 1.65 \cdot 10^{-14}]$. These bounds are propagated from the full original certificate, not new finite-series substitutions for the actual functions.

The actual original period satisfies $u_* \in [45.20902385898 \pm 3.31 \cdot 10^{-12}]$. In the u coordinate, multiply the four constants in RPE6 by $u_*^{10}, u_*^{16}, u_*^{16}, u_*^{16}$ respectively, from $z - z_* = -(u - u_*)/(uu_*)$. At $s = 1$ they are enclosed respectively by

$$[26954700 \pm 55.4], \quad [1.686109 \cdot 10^{10} \pm 7850], \quad [2.38964 \cdot 10^{10} \pm 10900], \quad [6677480 \pm 2.02]. \quad (\text{RPE9})$$

3.2 All four original labels on the old and corrected source spaces

The literal original marking remains

$$\frac{a}{\rho_a} \left| \begin{array}{cccc} \text{M} & \text{N} & \text{T} & \text{E} \\ 3/4 + 3i & 3/4 - 3i & 1/4 + 3i & 1/4 - 3i \end{array} \right., \quad \ell_a(S) = \prod_{b \neq a} \frac{S - \rho_b}{\rho_a - \rho_b} = \sum_{r=0}^3 \lambda_{ra} S^r. \quad (\text{RPE10})$$

Define $L = (\lambda_{ra}) = (V_\rho^\top)^{-1}$, as in FC26–35. Every coefficient is specified by the displayed product; in particular

$$\begin{aligned} (\lambda_{3M}, \lambda_{3N}, \lambda_{3T}, \lambda_{3E}) &= \frac{1}{435}(-24 - 2i, -24 + 2i, 24 - 2i, 24 + 2i), \\ \lambda_{2a} &= (\rho_a - 2)\lambda_{3a}, \quad \lambda_{1a} = (\rho_a^2 - 2\rho_a + 155/8)\lambda_{3a}, \\ \lambda_{0a} &= (\rho_a^3 - 2\rho_a^2 + (155/8)\rho_a - 147/8)\lambda_{3a}. \end{aligned} \quad (\text{RPE11})$$

These formulas follow by dividing the full root polynomial $\prod_a (S - \rho_a)$ by its factor $S - \rho_a$ and retaining the original derivative denominator.

On the old degree-three source, the four outputs at this period are

$$y_a^{\text{old}} = T_A \ell_a = \lambda_{2a} \mu_2 + \lambda_{3a} (\mu_3 + 3\mu_2 S'), \quad Y^{\text{old}} = \begin{pmatrix} 0 & 0 & \mu_2 & \mu_3 \\ 0 & 0 & 0 & 3\mu_2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} L. \quad (\text{RPE12})$$

The rank is exactly two and the kernel on that original source is $\mathcal{P}_1$. In label coordinates this kernel is exactly $\text{span}\{\mathbf{1}, \rho\}$, since interpolation sends those two evaluation vectors to $1, S$. In the original ES coordinates the kernel of $Y^{\text{old}} K^{-1}$ is therefore $K \text{span}\{\mathbf{1}, \rho\}$, with the full unchanged K of FC15a. No four-label column is removed or identified.

The actual order-two source, with its complete original quotient norm, is instead $\mathcal{U}_2 = \mathcal{P}_5/\mathcal{P}_1$. Its four retained inputs and outputs are

$$x_a^{(2)} = [S^2 \ell_a(S)], \quad y_a^{(2)} = T_A(S^2 \ell_a), \quad Y^{(2)} = \begin{pmatrix} \mu_2 & \mu_3 & \mu_4 & \mu_5 \\ 0 & 3\mu_2 & 4\mu_3 & 5\mu_4 \\ 0 & 0 & 6\mu_2 & 10\mu_3 \\ 0 & 0 & 0 & 10\mu_2 \end{pmatrix} L. \quad (\text{RPE13})$$

For its proof use the complete original Taylor translation formula

$$T_A S^{2+r} = \sum_{k=0}^r \binom{2+r}{k} \mu_{2+r-k} S'^k.$$

This evaluates every entry with the actual moments RPE2–3. Here

$$\det V_\rho^\top = -1305/4, \quad \det Y^{(2)} = \frac{180\mu_2^4}{-1305/4} = -\frac{16}{29}\mu_2^4 \in [-13191650 \pm 4.20] \neq 0. \quad (\text{RPE14})$$

Thus the corrected four-column map is exactly invertible, with all four original labels and the actual positive μ_2 .

There is also an exact operator identity giving every correction to the old map in RPE12. Define the unchanged derived symbols $T_A^{[r]} P(S') = \sum_{u,w} a_{uw} \beta_{8,u,w}^r P(S' + \beta_{8,u,w})$, for $r = 1, 2$. Expansion of $(S' + \beta_{8,u,w})^2$ inside the original sum proves

$$T_A(S^2 P) = S'^2 T_A P + 2S' T_A^{[1]} P + T_A^{[2]} P. \quad (\text{RPE15})$$

It retains all original coefficient and frequency contributions. The complete matrices of the last two maps on the old monomial source basis $(1, S, S^2, S^3)$ are

$$T_A^{[1]} = \begin{pmatrix} 0 & \mu_2 & \mu_3 & \mu_4 \\ 0 & 0 & 2\mu_2 & 3\mu_3 \\ 0 & 0 & 0 & 3\mu_2 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad T_A^{[2]} = \begin{pmatrix} \mu_2 & \mu_3 & \mu_4 & \mu_5 \\ 0 & \mu_2 & 2\mu_3 & 3\mu_4 \\ 0 & 0 & \mu_2 & 3\mu_3 \\ 0 & 0 & 0 & \mu_2 \end{pmatrix}. \quad (\text{RPE16})$$

Multiplying their coefficient matrices by L and applying RPE15 recovers RPE13 entry by entry. In particular the lost two old directions are restored by specified original moment terms, not by discarding the original kernel or choosing a different amplitude.

The source map $P \mapsto [S^2 P]$ is an isomorphism of the underlying four-dimensional spaces, with inverse obtained from the unique representative containing only degrees 2 through 5 and division by S^2 . Its exact norm is the pullback of the stated Gamma quotient norm; it is not asserted isometric to the old degree-three norm. Writing $h_{n,s} = M_s n!(s/2)_n$, the full Gram determinants are

$$\det G_{x,2} = \frac{h_{2,s} h_{3,s} h_{4,s} h_{5,s}}{(1305/4)^2}, \quad \det G_{y,2} = \frac{180^2 \mu_2^8 h_{0,s} h_{1,s} h_{2,s} h_{3,s}}{(1305/4)^2}, \quad \frac{\det G_{y,2}}{\det G_{x,2}} = \left(\frac{\mu_2}{2}\right)^8 \prod_{r=0}^3 \frac{\rho_{r+2}^2}{\rho_r^2}. \quad (\text{RPE17})$$

These follow from the unchanged leading coefficients of the original orthonormal polynomials, exactly as in FC28–30. Both Gram determinants retain the entire mass M_s^4 .

Finally define $X_2 e_a = x_a^{(2)}$, $Y_2 e_a = y_a^{(2)}$ and retain $\Phi_2 = X_2 K^{-1}$, $\Psi_2 = Y_2 K^{-1}$. The full original four-label intertwining is

$$T_A \Phi_2 = \Psi_2, \quad P_{\mathcal{U}_2} = \Phi_2 P \Phi_2^{-1}, \quad P_{\mathcal{W}} = \Psi_2 P \Psi_2^{-1}, \quad T_A P_{\mathcal{U}_2} = P_{\mathcal{W}} T_A, \quad (\text{RPE18})$$

with the complete original P from FC31, and $P_{\mathcal{W}}(y_a^{(2)}) = 2i y_M^{(2)}$ for every $a = \text{M, N, T, E}$. Its inverse matrices are the adjugates divided by RPE14 and FC15a's nonzero determinants; the pulled-back metrics are $(K^{-1})^* G_{x,2} K^{-1}$ and $(K^{-1})^* G_{y,2} K^{-1}$. Thus the actual real-period degeneration, its full fixed-space poles, and its exact order-two four-label correction are retained together.

3.3 The two lost scalars, their complete metric, and the total return

The rank-two old receiver has four distinct marked outputs. Indeed, in the ordered pair list (MN, MT, ME, NT, NE, TE), RPE11 gives

$$\lambda_{3a} - \lambda_{3b} = \frac{1}{435}(-4i, -48, -48 - 4i, -48 + 4i, -48, -4i)_{ab}. \quad (\text{RPE19})$$

The coefficient of S' in $y_a^{\text{old}} - y_b^{\text{old}}$ is $3\mu_2(\lambda_{3a} - \lambda_{3b}) \neq 0$. This proves all six distinctions without identifying a label with a conjugate label. For the exact metric statement, put $b = s/2$, $M = (2\pi)^{s/2}$, $c = a_{\text{amp}}/2$ and $c' = c - 4$. If this difference is $d_0 + d_1 S'$, its original squared norm is $M(|d_0 + c'd_1|^2 + b|d_1|^2) > 0$.

Let $\Phi_0 q = \sum_a (K^{-1}q)_a \ell_a(S)$ and $J_0 = T_A \Phi_0$. Retain the literal two coefficient functionals $t_r(q) = [S^r] \Phi_0 q$, $r = 0, 1$. Define the complete linear map

$$\Theta_2 : \mathbb{C}^4 \longrightarrow \mathcal{P}_1 \oplus \mathbb{C}^2, \quad \Theta_2 q = (J_0 q, t_0(q), t_1(q)). \quad (\text{RPE20})$$

Here the first $\mathcal{P}_1$ uses the original target variable S' ; the two scalar coordinates are the coefficients lost by the original degree-three conductor. For $Y = Y_0 + Y_1 S'$, put

$$c_3 = \frac{Y_1}{3\mu_2}, \quad c_2 = \frac{Y_0 - \mu_3 c_3}{\mu_2}, \quad p_Y(S) = c_2 S^2 + c_3 S^3, \quad (\text{RPE21})$$

$$\Theta_2^{-1}(Y, t_0, t_1) = K(p_Y(\rho_a) + t_0 + t_1 \rho_a)_a.$$

The two coefficients of $T_A p_Y$ are precisely Y_0, Y_1 by RPE12, whereas p_Y itself has zero constant and linear coefficients. These identities prove $\Theta_2 \Theta_2^{-1} = I$. Conversely any polynomial in the old source differs from its corresponding p_Y by exactly its original constant and linear terms; interpolation proves the other composition. In coefficient bases the determinant of Θ_2 is $3\mu_2^2 / ((\det V_\rho^T)(\det K)) \neq 0$. Thus both lost scalars, including their original ES coordinates, are recovered by finite displayed formulas.

Every cross term of the original source metric is as follows. Set $m_4 = b(3b + 2)$ and $m_6 = b(15b^2 + 30b + 16)$, and define

$$Q = \begin{pmatrix} 1 & c \\ c & c^2 + b \end{pmatrix}, \quad U = \begin{pmatrix} c^2 - b & c^3 - 3cb \\ c^3 + cb & c^4 - m_4 \end{pmatrix}, \quad (\text{RPE22})$$

$$H = \begin{pmatrix} c^4 + 2c^2b + m_4 & c(c^4 + 2c^2b + m_4) \\ c(c^4 + 2c^2b + m_4) & c^6 + 3c^4b + 3c^2m_4 + m_6 \end{pmatrix}.$$

With $v = (c_2, c_3)^\top$ and $t = (t_0, t_1)^\top$, direct integration against the unchanged WCF2 measure gives

$$\begin{aligned} \|p_Y + t_0 + t_1 S\|_{s,c}^2 &= M\{v^* H v + 2 \operatorname{Re}(t^* U v) + t^* Q t\} \\ &= M v^* (H - U^* Q^{-1} U) v + M(t + Q^{-1} U v)^* Q(t + Q^{-1} U v), \\ Q^{-1} U v &= \begin{pmatrix} -(c^2 + b)c_2 + (-2c^3 + 2c)c_3 \\ 2cc_2 + (3c^2 - 3b - 2)c_3 \end{pmatrix}, \quad (\text{RPE23}) \\ H - U^* Q^{-1} U &= \begin{pmatrix} 2b(b+1) & 6cb(b+1) \\ 6cb(b+1) & 18c^2b(b+1) + 6b(b+1)(b+2) \end{pmatrix}. \end{aligned}$$

For completeness, WCF2's recurrence gives the unchanged measure's moments M, Mb, Mm_4, Mm_6 in even degrees zero through six. One may verify them from $p_2 = y^2 - b$, $p_3 = y^3 - (3b + 2)y$ and their exact WCF2 norms. Odd moments vanish. Substitution of $S = c + iy$ then gives every entry of RPE22; for example $M^{-1}\langle S, S^3 \rangle = c^4 - m_4$ and $M^{-1}\langle S^2, S^3 \rangle = c(c^4 + 2c^2b + m_4)$. The inverse of Q is $b^{-1} \begin{pmatrix} c^2+b & -c \\ -c & 1 \end{pmatrix}$. Multiplication gives the last two expressions in RPE23 and

proves the completed square. The original quotient norm is its first term, attained at the exact two-scalar value $t = -Q^{-1}Uv$. No source mass, center, kernel direction or cross term is omitted.

For a full Gram directly in the displayed augmented coordinates, define

$$D_Y = \begin{pmatrix} 1/\mu_2 & -\mu_3/(3\mu_2^2) \\ 0 & 1/(3\mu_2) \end{pmatrix}, \quad G_{\Theta_2} = M \begin{pmatrix} D_Y^* H D_Y & D_Y^* U^* \\ U D_Y & Q \end{pmatrix}. \quad (\text{RPE24})$$

Thus $v = D_Y(Y_0, Y_1)^\top$ and RPE24 is exactly the original source metric transported by RPE21, in the order (Y_0, Y_1, t_0, t_1) . Its quotient block is $M D_Y^*(H - U^* Q^{-1} U) D_Y$. The target norm stays $\|Y\|_{s,c'}^2 = M(|Y_0 + c'Y_1|^2 + b|Y_1|^2)$; it is not replaced by that source quotient metric.

The four marked total states and their full common images retain both scalars:

$$\begin{aligned} \Theta_2(Ke_a) &= (y_a^{\text{old}}, \lambda_{0a}, \lambda_{1a}), \\ \Theta_2 P(Ke_a) &= (2iy_M^{\text{old}}, 2i\lambda_{0M}, 2i\lambda_{1M}). \end{aligned} \quad (\text{RPE25})$$

All constants on the right are RPE11–12's exact original values. Finally the entire polynomial return is

$$\widehat{P}_2(Y, t_0, t_1) = \Theta_2 P(K(p_Y(\rho_a) + t_0 + t_1 \rho_a)_a), \quad \Theta_2 P = \widehat{P}_2 \Theta_2, \quad \det D\widehat{P}_2 = -2. \quad (\text{RPE26})$$

The full original P is FC31, so these are explicit finite polynomial expressions with inverse coordinate map RPE21. The chain rule proves the determinant because the two constant determinants of Θ_2 and its inverse cancel. This uses the existing FC31 Jacobian identity with its ES–Fable provenance; it asserts no new counterexample beyond that source polynomial. Equations RPE20–26 retain the old map with both lost scalars; RPE13–18 separately give the actual order-two conductor on its correct quotient. The exact morphisms and their different original metrics have both been computed. No assertion places the chosen geometric quartet or unit on the arithmetic Riemann-zeta locus.

4 Every original cutoff: the complete two-pole law and its exact resonance

This section calculates every cutoff of the same original conductor at the real pair of RPZ1–19. It also identifies the entire resonance locus of the RPCJ simple real-pair family. No moment, original weight or higher coefficient is removed. The full geometric family and Gamma dictionary retain [5, 7, 2].

Retain RPCJ5–12's actual constants $c > 0$, $d = -iR$, $e < 0$, c_3, d_2, e_1, f . For each integer $D \geq 1$ set $N = D + 1$ and

$$B_{D,s}(z) = \text{diag}(\rho_0, \dots, \rho_D)^{-1} T_D(g(\cdot, z)) \text{diag}(\rho_0, \dots, \rho_D), \quad \rho_j = \sqrt{j!/(s/2)_j}. \quad (\text{RPAC1})$$

Here $T_D(g)_{ij} = g_{j-i}$ for $j \geq i$, and is zero otherwise; $g(t, z) = e^{-4\omega(t)} E_A(\omega(t), z)$ and $\omega(t) = -i \arctan t$, exactly RPCJ9. Both original values $s \in \{1, a_{\text{amp}}\}$ retain their full mass $(2\pi)^{s/2}$ and centers $a_{\text{amp}}/2$, $a_{\text{amp}}/2 - 4$. The displayed matrix uses their orthonormal frames.

Put $\tau = z - z_*$. Since $g_0 = c\tau^2 + O(\tau^3)$, it is nonzero in a sufficiently small punctured neighborhood of zero. There the exact inverse is the weighted triangular matrix of the coefficients $h_n = [t^n]g(t, z)^{-1}$, $0 \leq n \leq D$. Define the finite expressions

$$\begin{aligned} H_n &= \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^{n-j} \binom{n-j}{j} \frac{d^{n-2j} e^j}{c^{n-j+1}}, \quad H_{-1} = 0, \\ K_n &= -[y^n] \frac{c_3 + d_2 y + e_1 y^2 + f y^3}{(c + d y + e y^2)^2}. \end{aligned} \quad (\text{RPAC2})$$

Coefficient extraction in the second line is finite: explicitly it is $-\sum_{j=0}^{\min(3,n)} a_j \sum_{k=0}^{n-j} H_k H_{n-j-k}$, where $(a_0, a_1, a_2, a_3) = (c_3, d_2, e_1, f)$. Every constant on the right is an original derivative in RPCJ5–11. They are not independently prescribed values.

The complete inverse entries have the Laurent expansions

$$h_n(\tau) = \tau^{-n-2} \{H_n + \tau K_n + O(\tau^2)\}, \quad cH_0 = 1, \quad cH_n + dH_{n-1} + eH_{n-2} = 0 \quad (n \geq 1), \quad (\text{RPAC3})$$

where $H_{-2} = 0$. To prove both claims, substitute $t = \tau y$ as an exact auxiliary coefficient identity, retaining the inverse substitution and all powers:

$$\begin{aligned} g(\tau y, z_* + \tau) &= \tau^2(c + dy + ey^2) + \tau^3(c_3 + d_2y + e_1y^2 + fy^3) + O(\tau^4), \\ \tau^2 g(\tau y, z_* + \tau)^{-1} &= \frac{1}{c + dy + ey^2} - \tau \frac{c_3 + d_2y + e_1y^2 + fy^3}{(c + dy + ey^2)^2} + O(\tau^2). \end{aligned} \quad (\text{RPAC4})$$

These identities hold coefficientwise through each fixed degree D ; analyticity gives the stated remainder for every such finite set. Terms $g_j t^j$ with $j \geq 4$ first contribute at order at least four in the first line. They are retained in the remainder and in the exact inverse. Taking $[y^n]$ in the second line multiplies the original h_n by exactly τ^{n+2} , proving RPAC3. Expanding the reciprocal geometric series proves RPAC2, and multiplying it by $c + dy + ey^2$ proves the recurrence.

4.1 Exact resonance and the impossibility of consecutive zeros

In terms of the actual derivatives $A = G_z$ and $B = G_\xi$, put

$$\lambda = \frac{|B|}{|A|} > 0, \quad \theta = \frac{\Re(A\bar{B})}{|A||B|} \in [-1, 1]. \quad (\text{RPAC5})$$

Their magnitudes remain in every formula through $c = |A|^2$, $d = -2ic\lambda\theta$, $e = -c\lambda^2$. Define $U_0(X) = 1$, $U_1(X) = 2X$ and $U_n(X) = 2XU_{n-1}(X) - U_{n-2}(X)$. These are the explicitly defined polynomials; the familiar Chebyshev notation introduces no external bound. Induction in RPAC3 proves

$$H_n = \frac{(i\lambda)^n}{c} U_n(\theta), \quad H_{n-1}^2 - H_n H_{n-2} = \frac{e^{n-1}}{c^{n+1}} \quad (n \geq 1). \quad (\text{RPAC6})$$

For the second identity, its left side at $n = 1$ is c^{-2} . Substitution of the recurrence makes the left side at $n + 1$ equal to e/c times its value at n , proving it for all n . In particular H_D and H_{D-1} cannot vanish together.

The angle addition formula for sine verifies the same recurrence for $\sin((n+1)\phi)/\sin\phi$, with the same initial terms. The endpoints give $U_n(1) = n+1$ and $U_n(-1) = (-1)^n(n+1)$ by the recurrence. It follows that the complete zero set is

$$H_D = 0 \iff \theta = \cos \frac{k\pi}{D+1} \text{ for some } k = 1, \dots, D. \quad (\text{RPAC7})$$

Indeed these are D distinct roots of the degree- D polynomial, whose leading coefficient is 2^D , so there are no other roots. This characterizes all exceptional cutoffs without discarding any original derivative or asserting that a finite interval decides infinitely many equalities.

4.2 The entire limiting forward block and all exterior constants

For $D \geq 2$ the limiting matrix has nonzero block $C_{D,s}$ with rows $0, \dots, D-2$ and columns $2, \dots, D$:

$$(C_{D,s})_{ij} = \begin{cases} g_{j+2-i}(z_*) \rho_{j+2}/\rho_i, & j \geq i, \\ 0, & j < i, \end{cases} \quad 0 \leq i, j \leq D-2. \quad (\text{RPAC8})$$

Its diagonal is $e\rho_{i+2}/\rho_i$, so it is invertible. All higher coefficients $g_3, \dots, g_D$, with their actual moments and center corrections, remain in this block. For $D = 1$ use the empty block, whose zeroth exterior norm and determinant are one. The limiting forward rank is exactly $N - 2$. For $2 \leq r \leq N$ set

$$L_{r,D,s} = \frac{\|\bigwedge^{N-r} C_{D,s}\|}{c^N} > 0. \quad (\text{RPAC9})$$

The norms in RPAC9 are completely specified finite matrix norms of RPAC8 in the original orthonormal frames. In particular

$$L_{2,D,s} = \frac{|e|^{D-1} \rho_{D-1} \rho_D}{c^{D+1} \rho_0 \rho_1}, \quad L_{N,D,s} = c^{-N}. \quad (\text{RPAC10})$$

The first formula follows by multiplying the diagonal of RPAC8; its weight product telescopes to the displayed original ratios. It also gives c^{-2} at $D = 1$.

For every original cutoff and every $2 \leq r \leq N$ one has

$$\lim_{z \rightarrow z_*} |z - z_*|^{2N} \left\| \bigwedge^r B_{D,s}(z)^{-1} \right\| = L_{r,D,s}. \quad (\text{RPAC11})$$

To prove this identity, let the forward singular values of an invertible N -dimensional matrix be $\sigma_1 \geq \dots \geq \sigma_N > 0$. The product of the largest r inverse singular values is $\sigma_1 \dots \sigma_{N-r} / |\det B|$. Thus its r th inverse exterior norm equals $\|\bigwedge^{N-r} B\| / |\det B|$. Here $\det B = g_0^N$ and $g_0 = c\tau^2 + O(\tau^3)$. The limiting numerator is the strictly positive numerator in RPAC9. Continuity gives RPAC11, for complex as well as real approaches. This proves the exponent $2N$ at every exterior rank from two through N , without a resonance restriction.

4.3 Both complete inverse spectra

When $H_D \neq 0$, RPAC3 shows that the only entry of pole order $D + 2$ is the top-right entry, with coefficient $H_D \rho_D / \rho_0$. All other entries have lower pole order. Consequently, with $A_{D,s} = |H_D| \rho_D / \rho_0 > 0$, the two divergent inverse singular values have exact exponents and leading constants

$$(D + 2, D), \quad (A_{D,s}, L_{2,D,s} / A_{D,s}). \quad (\text{RPAC12})$$

The first constant follows by taking the norm after multiplication by τ^{D+2} . The second follows by dividing RPAC11 at rank two by the first singular asymptotic. Every other inverse singular value has a finite positive limit, the reciprocal of the nonzero forward singular values of $C_{D,s}$ in reverse order.

When $H_D = 0$, RPAC6 gives $H_{D-1} \neq 0$. The entire leading inverse matrix at order $D + 1$ has the block

$$N_{D,s} = \begin{pmatrix} H_{D-1} \rho_{D-1} / \rho_0 & K_D \rho_D / \rho_0 \\ 0 & H_{D-1} \rho_D / \rho_1 \end{pmatrix}. \quad (\text{RPAC13})$$

It occupies rows $0, 1$ and columns $D - 1, D$. All other entries vanish after multiplication by τ^{D+1} . For $D = 1$ this is the full matrix, with diagonal entries $H_0 = 1/c$; hence the statement also covers the endpoint. The block is invertible. Its two positive singular values are the exact leading constants of the two divergent inverse singular values, both of exponent $D + 1$. For an explicit finite formula write its entries as $\begin{pmatrix} a & b \\ 0 & d_0 \end{pmatrix}$, set $S = |a|^2 + |b|^2 + |d_0|^2$ and $Q = |ad_0|^2$, and obtain

$$\sigma_{\pm}(N_{D,s}) = \left(\frac{S \pm \sqrt{S^2 - 4Q}}{2} \right)^{1/2}. \quad (\text{RPAC14})$$

This follows from the trace and determinant of $N_{D,s}^* N_{D,s}$. The full K_D in RPAC2 retains every next-jet correction. Furthermore RPAC6 at $n = D$ gives $|H_{D-1}|^2 = |e|^{D-1} / c^{D+1}$, so $|\det N_{D,s}| = L_{2,D,s}$, consistently with RPAC11.

These calculations exhaust the alternatives. The inverse exterior exponents are

$$\begin{cases} (D+2, 2N, \dots, 2N), & H_D \neq 0, \\ (D+1, 2N, \dots, 2N), & H_D = 0. \end{cases}$$

The first exterior constants are $A_{D,s}$ and $\sigma_+(N_{D,s})$, respectively; all remaining constants are RPAC9. At cutoff $D = 0$ the single matrix entry is g_0 , so its sole inverse pole has exponent two and constant c^{-1} . In the original coordinate u , every constant at pole exponent p is multiplied by $|u_*|^{2p}$, by the exact identity $z - z_* = -(u - u_*)/(uu_*)$.

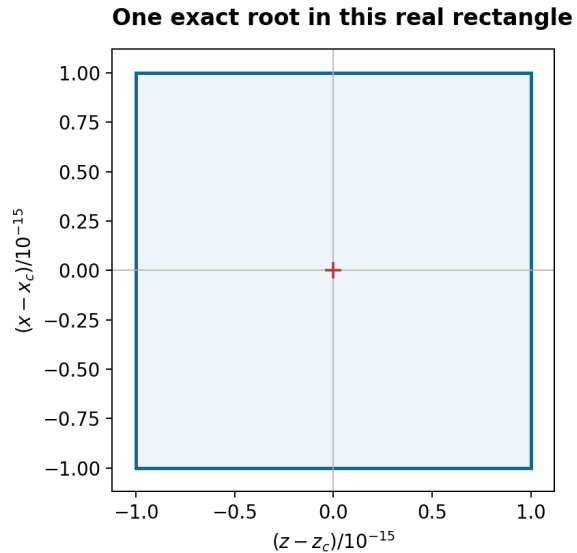
4.4 An actual finite range on the certified original pair

The companion `check_real_pair_all_cutoffs.py` first checks the finite coefficient and reciprocal identities exactly, then propagates the complete RPZ certificate's outward enclosures for c , $|B|^2$ and R to the actual $\theta = R/(2\sqrt{c|B|^2})$. For every $D = 1, \dots, 100$ it encloses each of the D values $\cos(k\pi/(D+1))$ and verifies strict separation from that same interval for θ . Every one of these 5050 comparisons is with an enclosing interval; no decimal equality test is used. The full receipt records the minimum gap and the source-certificate hash. It verifies

$$\left| \theta - \cos \frac{k\pi}{D+1} \right| > 1.95 \cdot 10^{-5} \quad (1 \leq k \leq D \leq 100). \quad (\text{RPAC15})$$

This strict bound proves $H_D \neq 0$ throughout those one hundred cutoffs. Therefore the actual original pair has the singular poles $(D+2, D, 0, \dots, 0)$ and exterior poles $(D+2, 2D+2, \dots, 2D+2)$ for every $1 \leq D \leq 100$, with every constant given by RPAC9–12. For larger cutoffs, RPAC2 and RPAC7–14 remain the exact complete formulas; a finite interval check does not assert nonresonance at every integer cutoff. The arithmetic underlying the outward comparisons is Johansson's Arb [8]; the polynomial recurrence, full pole law and its original-metric constants have been proved above.

A positive real period: exactly two conjugate factors vanish



Display offsets only; inverse coordinates:
 $z = z_c + 10^{-15}X$, $x = x_c + 10^{-15}Y$
 The cross marks the center, not an exact root position.

$z_c = 0.022119477808661845132007916399\dots$
 $x_c = -1.466817440463494183975448871845\dots$
 Exact terminating-decimal centers: RPZ4.

The full infinite series is certified

$$T(y) = y - PF(y), \quad F = (\Re d_1, \Im d_1)$$

$$T(X) \subset \text{int}(X), \quad \text{Lip}_r(T) < 1.829 \cdot 10^{-6}$$

Geometric iteration proves existence and uniqueness.

$$u_* = 1/z_* \in [45.20902385898 \pm 3.31 \cdot 10^{-12}]$$

$$a_* = \alpha(1 + ix_*), \quad \alpha \in \mathbb{R} \setminus \{0\}$$

$$d_1 = d_4 = 0, \quad d_2, d_3 \neq 0$$

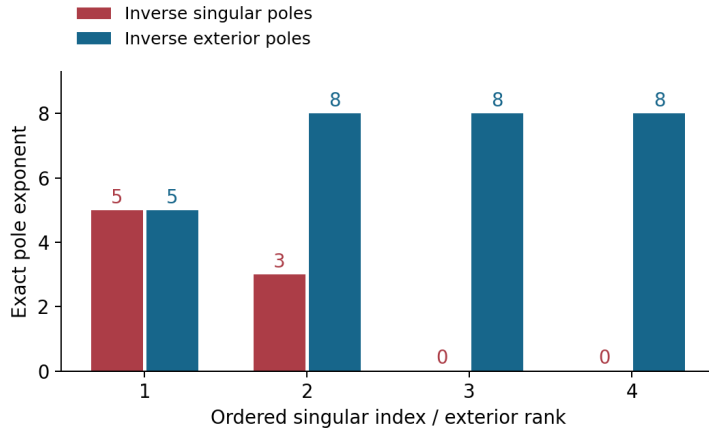
$$E_A = GG^\#, \quad G = f_1 f_2$$

$$\mu_0 = \mu_1 = 0, \quad \mu_2 = 2|G_\xi|^2 > 0$$

Proof: RPZ1–19 and RPCJ1–8. Full original matrix, unit and factor signs retained.
 Inclusion arithmetic: Fredrik Johansson, Arb, arXiv:1611.02831v1. This is a geometric coefficient member; no arithmetic zeta-zero identification.

Figure 1: The exact real rectangle RPZ4 contains one zero of the complete original factor, by RPZ5–14. The center marker is not the exact root. The axes are display offsets with the full inverse coordinate rule. The original unit and both conjugate pairs are retained; RPZ15–19 proves their exact orders. Original period: [5]. Inclusion arithmetic: Fredrik Johansson [8].

Degree three: the old inverse diverges; all four labels are retained



$$\tau = z - z_*, \quad \|\wedge^r B^{-1}\| \sim C_r |\tau|^{-p_r}$$

$$C_1 = |R(2cb - R^2)|\rho_3/(c^4\rho_0)$$

$$C_2 = \mu_2^2 \rho_2 \rho_3 / (4c^4 \rho_0 \rho_1)$$

$$C_3 = \|C_s\|/c^4, \quad C_4 = c^{-4}$$

All constants are strictly positive.
 Original period: each constant gains $u_i^{2p_i}$.
 Original Gamma mass, center and weights: RPCJ9; RPE5–9.

Exact exponents: RPCJ14–23. Four distinct old marks, two-scalar inverse and full metric: RPE10–26.
 Human sources: ES reader v69 (The Clankers; underlying Alpöge/Fable/Tao example), and the cited DLMF Chapter 18 authors for the Gamma dictionary.

Both exact ways to retain all four labels

Old source loses exactly two directions

$$T_A : \mathcal{P}_3 \longrightarrow \mathcal{P}_1, \quad \ker T_A = \mathcal{P}_1$$

$$\Theta_2 q = (T_A \Phi_0 q, [S^0] \Phi_0 q, [S^1] \Phi_0 q)$$

The two saved scalars give an exact inverse (RPE21).

The actual order-two conductor is invertible

$$T_A : \mathcal{P}_5/\mathcal{P}_1 \longrightarrow \mathcal{P}_3$$

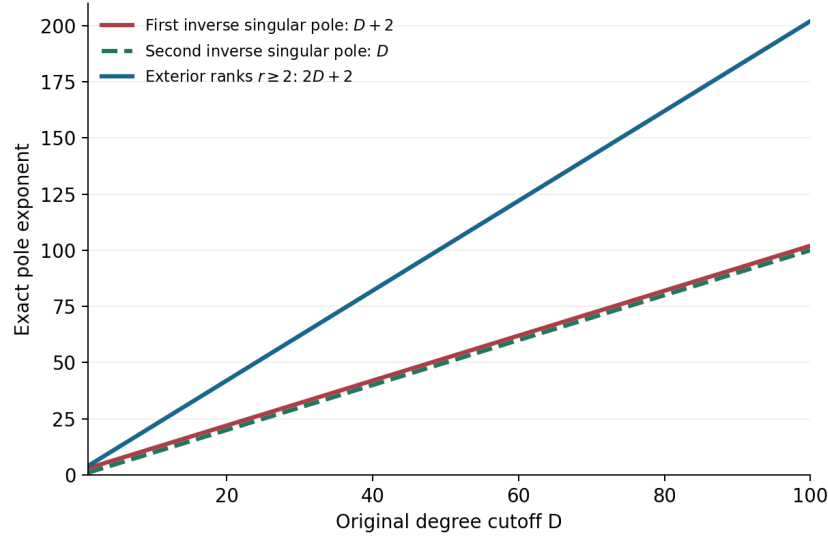
$$x_a = [S^2 \ell_a], \quad y_a = T_A x_a$$

$$a \in \{M, N, T, E\}$$

$$\det Y^{(2)} = -\frac{16}{29} \mu_2^4 \neq 0$$

Figure 2: RPCJ14–23 and RPE5–9 calculate the actual original Gamma-metric poles and every constant. RPE10–26 retains four distinct marks, the two lost old coordinates, their inverse and all source metric cross terms, as well as the actual order-two quotient. The displayed spaces are different exact domains of the same original conductor. ES marking: [3]; original Gamma dictionary and human authors: [7, 2].

The original real-pair inverse: every cutoff, both exact spectra



Each integer cutoff on this plot is certified.
 Lines join integer values; there is no noninteger-cutoff claim.
 Coefficients retain all original Gamma weights and moments.
 Exact constants: RPAC8-14.

The complete resonance equation

$$H_D = \frac{(i\lambda)^D}{c} U_D(\theta)$$

$$H_D = 0 \iff \theta = \cos \frac{k\pi}{D+1}$$

$$k = 1, \dots, D$$

$$\theta = \frac{\Re(A\bar{B})}{|A||B|}, \quad \lambda = \frac{|B|}{|A|}$$

At the certified original pair, for $1 \leq D \leq 100$:

$$|\theta - \cos(k\pi/(D+1))| > 1.95 \cdot 10^{-5}$$

All 5,050 interval comparisons pass.

For every D, the exact alternative on resonance is

$$(D+1, D+1, 0, \dots, 0)$$

Proof RPAC1-15. A finite certificate proves nonresonance through D=100; the all-degree formulas retain both cases.
 Original period and Gamma dictionary: PCL1-16 and WCF1-6. Inclusion arithmetic: Fredrik Johansson, Arb, arXiv:1611.02831v1.

Figure 3: RPAC1-14 proves the complete original-cutoff inverse law and both exact resonance alternatives, with all original moments and weights. RPAC15 rigorously excludes resonance at every integer cutoff from 1 through 100 for the actual certified real pair; the lines join those integer exponents. It makes no all-degree nonresonance assertion. Original construction: [5, 7]; inclusion arithmetic: Fredrik Johansson [8].

References

- [1] Split-Zero programme, accepted 782-page source collection, DFX4–5 and CNC1–2, consulted local source snapshot. The exact accepted source is included as `source_dependencies/ACCEPTED_782_CUMULATIVE.tex`. Its DFX4–5 occur at lines 30623–30633 and CNC1–2 at 38157–38178. No live public link to this exact source edition is asserted before publication of the included file.
- [2] T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, NIST Digital Library of Mathematical Functions, Chapter 18, <https://dlmf.nist.gov/18>, equations 18.19.8–9, 18.22.8 and 18.23.7. The original orthogonality and generating-function dictionary is credited in WCF1–6; RS21–26 applies it with the complete original centers, phases, Gamma mass and conductor inverse moments.
- [3] The Clankers, *Erdős–Straus project reader*, canonical 488-page edition, deposited source version 69, `erdos_straus_project_reader.tex`, theorem `explicit-four-time-Jacobian-collision`, source lines 19484–19588. Source SHA-256: 678b3f70ee7d2a6d00f6c588d7778e390118ab4743a5759612dbba6430984c78. The formal source uses the labels M,N,T,E. The seven-point fibre and missing eighth factor-sign state are proved here as an extension, rather than attributed to this reader. The complete original version-69 author source, with the stated hash and byline, is included as `source_dependencies/ES_READER_V69.tex`.
- [4] Split-Zero programme, *Vanishing of the original proper-source kernel through conductor coprimality*, 19 September 2026, NC1–5, `ORIGINAL_CONDUCTOR.COPRIMALITY.tex`.
Pinned public proof: NC1–5.
- [5] Split-Zero programme, *The literal original conductor limit and an invertible pivot restriction*, `PCL_COMPLETE.tex`, PCL1–16. Original supplied programme TeX; the entire-period coefficient formula and prior positivity slice are retained and cited at their use in OFM6–11. This is a programme source, not an attribution of that construction to the classical human sources.
Pinned public proof: PCL1–16.
- [6] Supplied Split-Zero programme, *An evaluated original-period domain with zero conductor vanishing order; the two exceptional unit phases; universal period receivers*, `PERIOD_ZERO_COMPLETE.PROOF.tex`, source collection 17 September 2026, PZ1–13, PZX1–6 and PZU1–11. No individual byline is supplied in this programme proof fragment. Exact source SHA-256: cbb4108de647f61af1e49deeeff46de97593eb9bfa9a6a33705dea344df3ec19. The unchanged complete source is included with the dependencies. PZ12 already proves the four-factor order bound and moment test; PZX supplies both inverse-period leading coefficients, and PZU the finite zero-free period domain. These prior results are credited at their use and are not claimed as new results of this continuation.
The full exact provider is included as `source_dependencies/PERIOD_ZERO_COMPLETE.PROOF.tex`; no unverified public target is substituted for that source.
- [7] Split-Zero programme, *A polynomial forward bound for the full weighted conductor*, `WEIGHTED_CONDUCTOR.FORWARD.tex`, WCF1–6, source intake 17 September 2026. This source credits the Meixner–Pollaczek dictionary to the NIST DLMF Chapter 18 by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, equations 18.19.8–9, 18.22.8 and 18.23.7; its original phases and masses are retained here, and no new orthogonality theorem is asserted.

Pinned public proof: WCF1–6.

- [8] Fredrik Johansson, *Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic*, original author source, arXiv:1611.02831v1 (9 November 2016), <https://arxiv.org/abs/1611.02831v1>. Source sections “Features and example applications”, “Radii and magnitude bounds” and “Precision and bounds” support the stated inclusion arithmetic. The complete FPZ series-tail and zero-count proofs are given here; they are not attributed to the software paper.